



इलाहाबाद विश्वविद्यालय
UNIVERSITY OF ALLAHABAD
(A Central University)



A PROJECT REPORT ON

MEDICAL INSURANCE COST PREDICTION

Submitted by: Komal Vishwakarma

Examination Roll No: 24421696

Under the Guidance of

Dr. G. Madhusudan Sir

University of Allahabad



इलाहाबाद विश्वविद्यालय UNIVERSITY OF ALLAHABAD (A Central University)



This is to certify that **Komal Vishwakarma**, Roll No:**23124325** student of **M.Sc. Statistics (4th semester)** Department of Statistics, University of Allahabad, carried out the Dissertation entitled "**Medical Insurance Cost Prediction**" in the partial fulfilment for the requirement of award for Master of Science degree in Statistics. The Dissertation has not formed the basis for the award of any degree, diploma, associate-ship, or fellowship. The Dissertation represents independent work carried out by the candidate.

Date:-

Signature:-

Supervisor:-

Dr.G. Madhusudan

University of Allahabad



इलाहाबाद विश्वविद्यालय UNIVERSITY OF ALLAHABAD

(A Central University)



I **Komal Vishwakarma** student of **M.Sc. Statistics (4th semester)**, **Department of Statistic,**
University of Allahabad, hereby declare that the Dissertation report entitled,

“Medical Insurance Cost prediction”, in part al fulfilment for the award of degree of Master of Science in statistics is a record of original work carried out by me under the supervision of **Dr. G.Madhusudan**
University of Allahabad. This project has not been submitted earlier in part or full for the award of any degree, diploma, associate ship or fellowship.

Date:

Place: Prayag Raj

Komal Vishwakarma

Roll No:23124325

Contents:

1. ABSTRACT
2. ACKNOWLEDGMENT
3. INTRODUCTION
4. AIM OF STUDY
5. MACHINE LEARNING
 - 5.1 Supervised
 - 5.2. Unsupervised
 - 5.3. Reinforcement
 - 5.4. Semi Supervised
6. METHODOLOGY
 - 6.1. Dataset Descript on
 - 6.2. Data Pre-processing
 - 6.2.1. Checking missing values
 - 6.2.2. Exploratory Data Analysis
 - 6.2.3. Visualization
7. Model Fitting
 - 7.1.1. Linear Regression
 - 7.1.2. Decision Tree Regression
 - 7.1.3. Random Forest Regression
 - 7.1.4. Support Vector Regression
 - 7.1.5. Gradient Boosting Regression
8. Conclusion

Abstract:

A policy that helps to cover all loss in terms of costs brought on by various hazards is insurance. The price of insurance is influenced by a number of factors. The expression of the cost of an insurance policy is influenced by these considerations of many aspects. The insurance industry can use machine learning (ML) to improve the efficiency of insurance. Machine learning (ML) is a well-known research field in the fields of computational and applied mathematics. When it comes to utilizing historical data, ML is one of the computational intelligence components that may be addressed in a variety of applications and systems. ML has several restrictions, so; In the healthcare sector, predicting medical insurance costs using ML techniques is still a challenge, necessitating further research and development. This project offers a computational intelligence method for forecasting healthcare insurance expenses using machine learning algorithms. Linear regression, Decision Tree regression, Gradient boosting regression, and are all used in the proposed study methodology. For the goal of cost prediction, we used a dataset of medical insurance costs that we downloaded from the KAGGLE repository. Machine learning techniques are used to demonstrate the forecasting of insurance costs by regression models and compare their degrees of accuracy.

Acknowledgment:

My sincere appreciation to my supervisor, Dr. G Madhusudan Sir. who gave me his incredible and priceless time to help me at every stage where I thought I needed it with my project. I also want to express my gratitude for his punctuality, which I have noticed during my project work.

I also want to appreciate the Statistics Department, which has always improved my academic

experience and given me a flexible studying environment. Thank we to every research scholar who has ever helped me and inspired me.

My sincere gratitude goes out to all my friends who helped me with my project work and helped keep me motivated.

Introduction:

Threats and uncertainty abound on the globe where we dwell. People, homes, durables, and assets are all subject to a variety of dangers, with varying degrees of risk. These dangers include everything from the chance of contracting illnesses to dying without protection and losing money or property. Risks, however, cannot always be avoided, thus the financial industry has created a wide range of products to protect people and businesses from these risks by employing money. Therefore, one of the strategies that either lowers or eliminates loss costs brought on by diverse risks is insurance. the importance of insurance in people's lives. It is crucial that insurance providers estimate the amount covered by this policy with appropriate accuracy.

The estimation of insurance costs is influenced by a number of elements or factors, each of which is significant. The overall cost of the insurance will alter if any component is added or removed when the amounts are determined. Therefore, it is important to do these jobs precisely. Because there is a significant likelihood of human error, insurance brokers also employ a variety of technologies to determine the insurance premium. Thus, ML is useful in this situation. The effort or procedure used to create the policy may be generalized using ML. These machine learning models are self-learning. The model is trained using historical insurance data. Using the components required to calculate the payments as its input, the model can then reliably anticipate the costs of insurance policies. Reduces the need for human resources and labour while increasing business profitability. As a result, ML can increase accuracy. Our objective is to estimate insurance prices. The cost of insurance premiums depends on a number of factors. Insurance costs are therefore ongoing. Regression is the greatest option for meeting our needs. Since there are several independent factors utilized to determine the dependent (target) variable, multiple linear regression is employed in this research. The dataset for health insurance costs is utilized in this investigation. First, the dataset underwent pre-processing. We then used training data to create regression models, and testing data to assess these models. Multiple linear regression, decision tree regression, and other regression models were utilized in this article. reduces the need for human resources and labour while increasing business profitability. Multiple linear regression, decision tree regression, and gradient boosting regression can all help to increase accuracy in this way. It is discovered that the gradient boosting method, with an r-squared value of 86.7853, offers the most accuracy. The major objective of this study is to incorporate a fresh way of estimating insurance costs.

Aim of the Study:

In this project, by using a set of ML algorithms, a computational intelligence approach is applied to predict healthcare insurance costs.

Why machine learning:

One of the most well-known branches of artificial intelligence is machine learning. Nearly every industry, including healthcare, finance, infrastructure, marketing, self-driving vehicles, recommendation systems, chatbots, social media, gaming, cyber security, and many more, uses machine learning techniques.

Machine learning is now in the development stage, and several new technologies are constantly being introduced. It benefits us in a variety of ways, including large-scale data analysis, data extraction, interpretation, etc. As a result, there are countless applications for machine learning.

What is Machine Learning?

Machine learning, a subfield of artificial intelligence, enables robots to autonomously learn from experience and advance. It is referred to be the area of research that enables computers to learn without explicit programming. In comparison to traditional programming, it is very different.

How Machine Learning Works?

A fundamental component of artificial intelligence is machine learning, which enables computers to learn from the past and predict the future. Data exploration and pattern matching are involved, with little assistance from humans. Machine learning has mostly been used to four technologies:

1. Supervised Learning:

A machine learning technique called "supervised learning" necessitates monitoring just like a Student teacher interaction does. In supervised learning, a computer is taught using data that has already been correctly labelled with some of the outputs. As a result, supervised learning algorithms analyze sample data whenever fresh data is entered into the system and use that labelled data to anticipate accurate results.

It falls under one of two categories for algorithms. These are listed below:

When output is presented as a category, such as Yellow, Blue, Right or Wrong, etc., this is referred to as classification.

Regression: It works with variables whose outputs have real values, such as age, height, etc. We can gather or create data output from experience thanks to technology.

2. Unsupervised Learning:

Unsupervised learning does not require categorized or clearly labelled data to train a system, in contrast to supervised learning. Even without any labelled training data, it seeks to create groups of unsorted data based on certain patterns and differences. Since there is no supervision in unsupervised learning, the machines are not given any sample data. As a result, robots can only discover hidden structures in unlabeled data.

It falls under one of two categories for algorithms. These are listed below:

Clustering: It deals with situations when the training data must be grouped inherently, such as when categorizing students according to their areas of interest.

Association: It addresses the guidelines that aid in identifying a sizable chunk of data, such as students interested in machine learning (ML) and intrusted in AI.

3. Semi-supervised learning:

Combining supervised and unsupervised learning techniques is known as semi-supervised learning. It is used to get around both supervised and unsupervised learning's shortcomings. In the semi-supervised learning approach, both labelled and unlabeled data are used to train the system. Nevertheless, it includes both a sizable number of unlabeled cases and a few examples with labels. Some of the most well-known semi-supervised Learning real-world applications are speech analysis, online content classification, protein sequence classification, and text document classifiers.

4. Reinforcement learning:

A feedback-based machine learning technique known as reinforcement learning is one that excludes the need for labelled data. Through performing actions and observing how they affect the environment; an agent learns how to behave there. Agents can provide compliments for each constructive activity and criticism for destructive ones. Since there are no training data for reinforcement learning, agents can only learn from their experience.

METHODOLOGIES

As per the aim of our problem, we will be working to predict whether the client has agreed to place a term deposit. To come out with a conclusion we will do the following steps:

- Analyzing data for missing value
- Exploratory Data Analysis
- Categorical Encoding
- Checking and Dealing with Multicollinearity
- Outlier detection and removal after standardization
- Data imbalance
- Model fitting and Comparison

DATASET

We collected the data set from the Kaggle website in order to build the claim cost model prediction. There are seven features in the data set. The data set is divided into two parts: training data and test data. Of the total data utilized, training data account for around 80%, and test data for the remaining 20%. The test set will be used to evaluate the regression model after the training data set has been used to develop a model as a predictor of medical insurance cost year. The description of the dataset is displayed in the following table.

The dataset has 1338 records in total. For a given person, each record includes the information below.

Age – Age of the person

Sex – Sex of the person

BMI – Body Mass Index (BMI) of the person

Children – Number of children for the person

Smoker – Smoking status of the person

Region – Region of the person in US

Charges – Medical Insurance costs per year for the person

Import essential libraries:

```
import numpy as np
import pandas as pd    import
matplotlib.pyplot as plt

import seaborn as sns
```

In this project, I utilized several libraries such as NumPy, pandas, matplotlib, seaborn, etc.

NumPy: NumPy is a library for the Python programming language that supports large, multidimensional arrays and matrices. It offers a wide range of high-level mathematical operations that can be performed on these arrays. Additionally, NumPy plays a crucial role as the foundation of the machine learning stack.

Pandas: Pandas is primarily used for working with Data Frames, a popular data structure in machine learning. It provides convenient tools for importing and exporting tabular data in various formats, including CSV and JSON files.

Matplotlib: Matplotlib is a well-known and widely used Python plotting library in the field of machine learning. It enables the creation of various visualizations, allowing us to better understand and interpret large amounts of data.

Seaborn: Seaborn is a library that facilitates data exploration and comprehension. Its charting functions are designed to work seamlessly with data frames and arrays containing complete datasets. Seaborn internally handles the necessary semantic mapping and statistical aggregation to generate meaningful and informative graphs.

DATA PRE-PROCESSING

The adjustments made to our data prior to feeding it to the algorithm are referred to as preprocessing. Data pre-processing is a method for transforming unclean data into clean data sets. In other words, anytime data is acquired from various sources, it is done so in a raw manner that makes analysis difficult.

Need of Data Pre-processing

- The format of the data in machine learning projects must be correct in order to get better results from the applied model. For example, the Random Forest algorithm does not tolerate null values, hence null values must be handled from the original raw data set in order to execute the Random Forest algorithm. Some specific Machine Learning models require data in a specific format.
- Another consideration is that the data set should be organised such that many Machine Learning.
- can run simultaneously and the best one is selected.

Checking Missing Value:

Use the `DataFrame.isnull()` function to determine if a value is null and the `sum()` method to compute the total.

```
data.isnull().sum() age      0 sex      0 bmi      0 children
0 smoker    0 region    0 charges  0
dtype: int64
```

There are no missing values as such,

This means we do not have to worry about imputation or dropping rows or columns with missing data.

Converting Categorical Features to Numerical Features:

The dataset contains three categorical features, Categorical features are data types represented as strings, which are readily understandable by humans. However, machines lack the ability to directly interpret categorical data. Hence, it becomes necessary to convert categorical data into numerical format for further analysis.

```
clean_data = {'sex': {'male': 0, 'female': 1},
              'smoker': {'no': 0, 'yes': 1},
              'region': {'northwest': 0, 'northeast': 1, 'southeast': 2, 'southwest': 3}
            }

data_copy = data.copy() data_copy.replace(clean_data,
inplace=True)
```

The levels of a categorical variable are excessive. The model's performance is lowered as a result..

Levels of a categorical variable are sporadic. There is little probability that many of these values will really affect how well the model fits. For instance, a variable 'illness' may have some levels that would only sometimes happen.

For the majority of the observations in the data set, there is just one level, which means that it always happens. Due to their extremely low variance, variables with such levels do not have a beneficial effect on model performance.

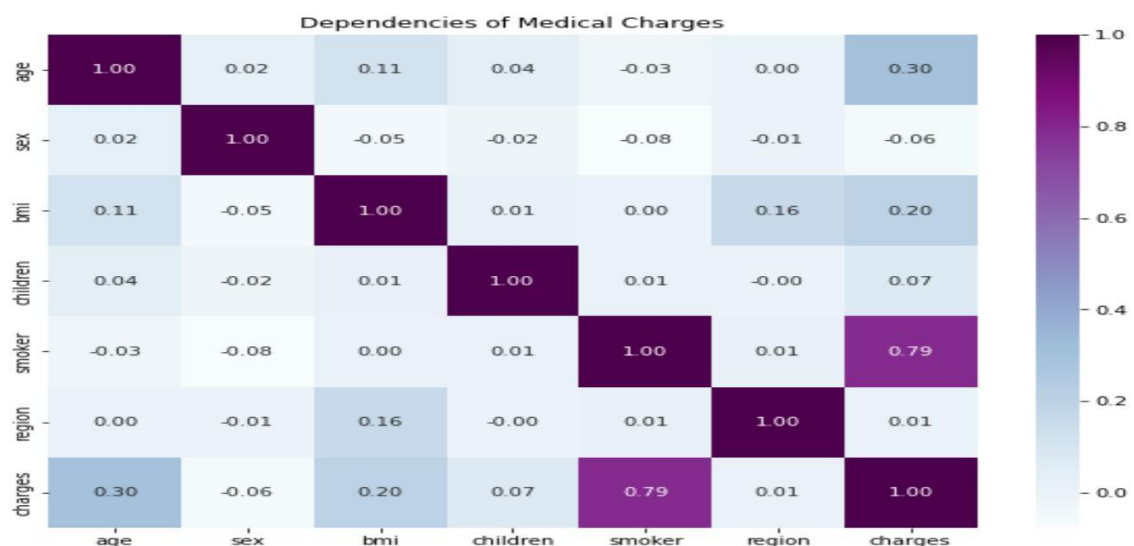
Data visualization:

The graphic display of information and data is known as data visualisation. Data visualisation tools offer an easy approach to observe and comprehend trends, outliers, and patterns in data by utilising visual components like charts, graphs, and maps. Additionally, it offers a great tool for staff members or business owners to clearly deliver data to non-technical audiences. Data visualisation is significant because it makes data easier to view, interact with, and comprehend. No of their degree of knowledge, the correct visualisation can put everyone on the same page, whether it is straight forward or complex.

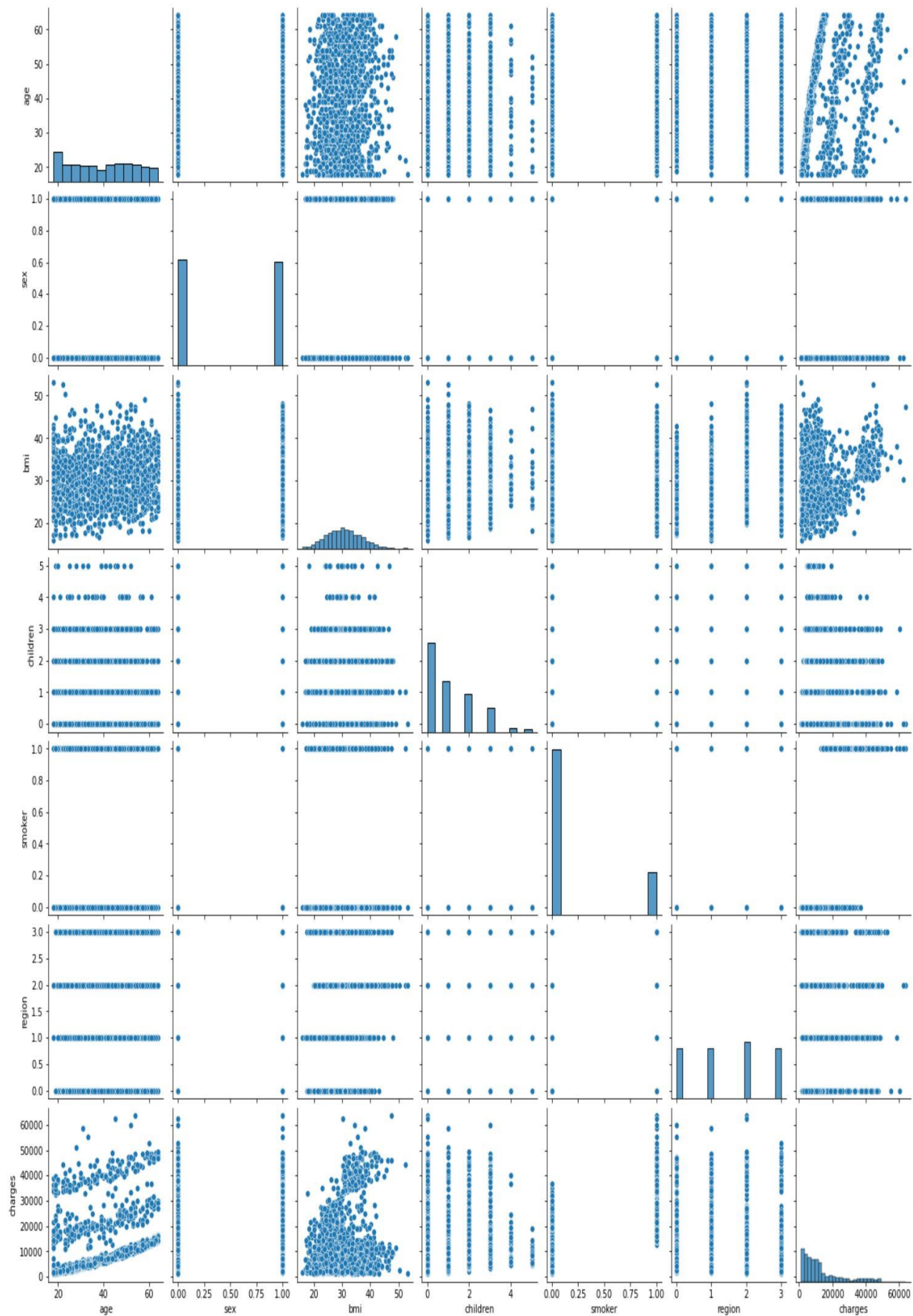
CORRELATION MATRIX

A correlation plot is a visual representation of the correlation matrix, a table that shows the correlation coefficients between pairs of variables in a dataset. The correlation coefficient measures the strength and direction of the linear relationship between two variables, with values ranging from -1 to +1.

```
corr = data_copy.corr()
fig, ax = plt.subplots(figsize=(10,8))
sns.heatmap(corr,cmap='BuPu',annot=True,fmt=".2f",ax=ax) plt.title("Dependencies of Medical Charges") plt.show()
```



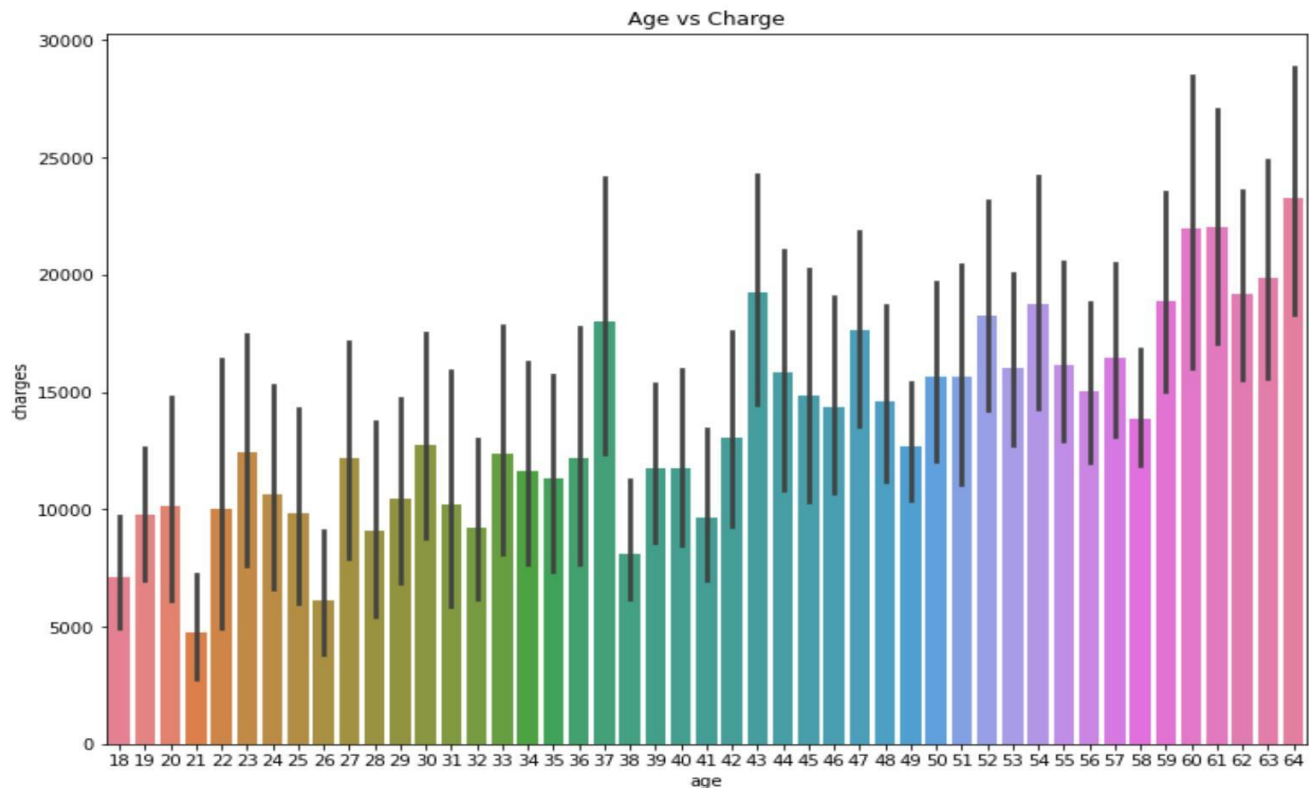
```
plt.figure(figsize=(6,6))
sns.pairplot(data_copy)
plt.show()
```



Age vs. Charges

```
plt.figure(figsize=(12,9)) plt.title('Age  
vs Charge')  
sns.barplot(x='age',y='charges',data=d  
ata_copy,palette='husl')
```

Out [14] : <AxesSubplot: title={'center': 'Age vs Charge'}, xlabel='age', ylabel='charges'>



We can see from the graph that as people age, their insurance costs will rise. As indicated in Figure, the insurance fee, for instance, is 23,000 when the age reaches 64. Charges are listed on the y-axis, while age is displayed on the x-axis.

Region vs charges:-

```
plt.figure(figsize=(10,7)) plt.title('Region  
vs Charge')  
sns.barplot(x='region',y='charges',data=data_copy,palette='Set3')
```

Out [15] : <AxesSubplot: title={'center': 'Region vs Charge'}, xlabel='region', ylabel='charges'>

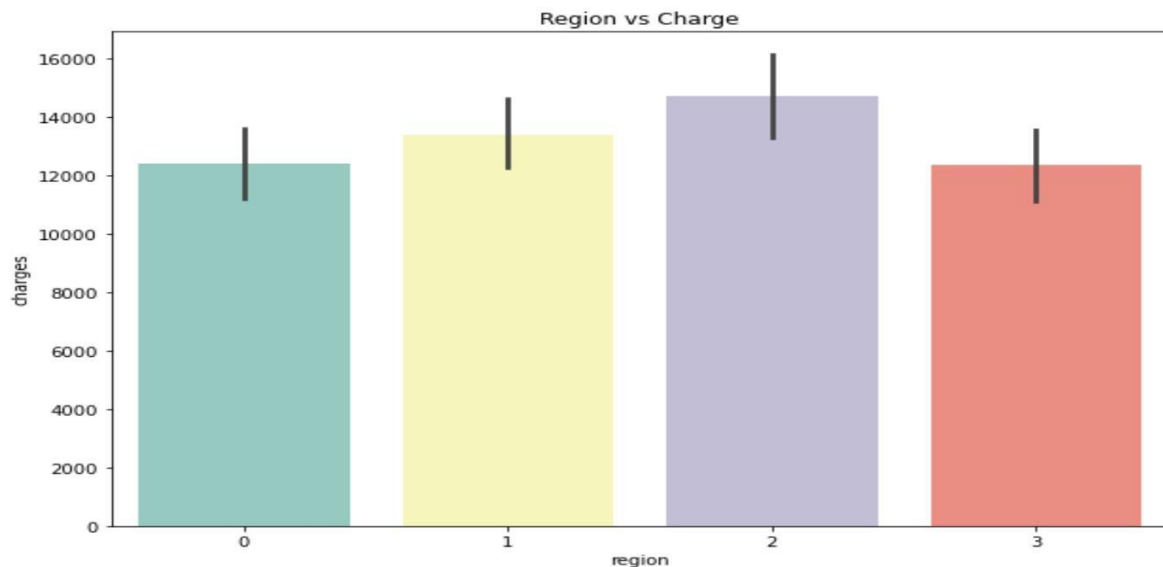
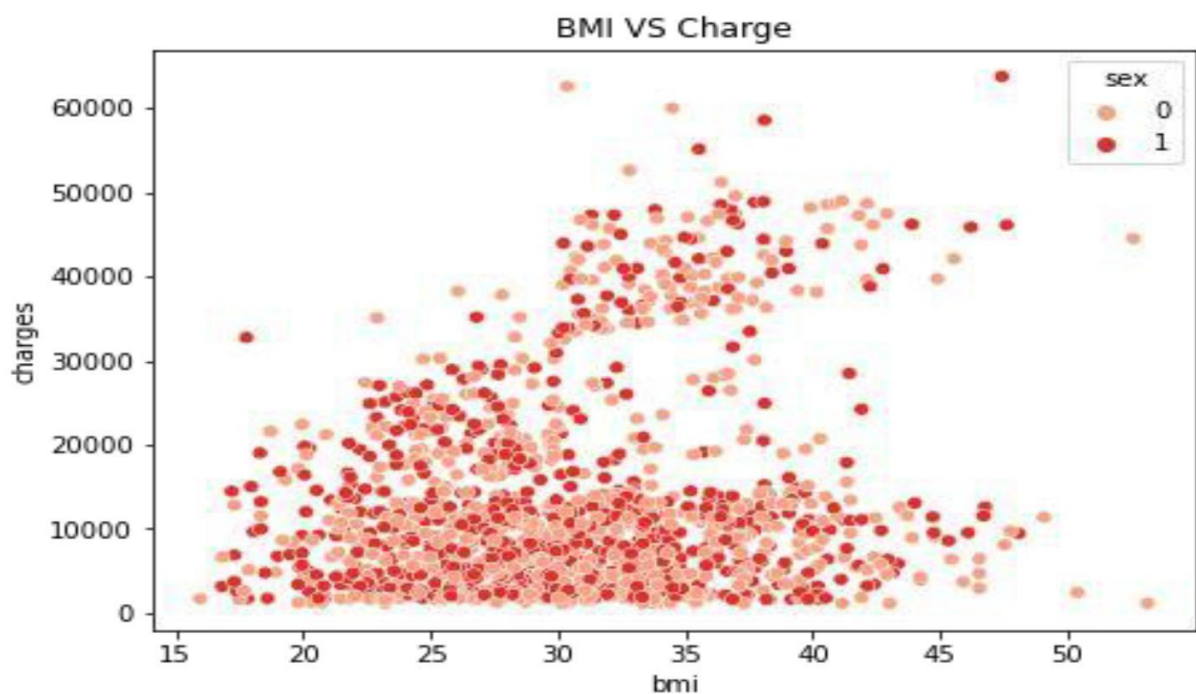


Figure illustrates how insurance costs differ by area. In comparison to other regions, the southeast has higher health insurance costs. Charges are indicated on the y-axis, while the area is displayed on the x-axis.

BMI vs Charges

```
plt.figure(figsize=(7,5))
sns.scatterplot(x='bmi',y='charges',hue='sex',data=data_copy,palette='Reds') plt.title('BMI VS Charge')
```

Out[16]: Text(0.5, 1.0, 'BMI VS Charge')

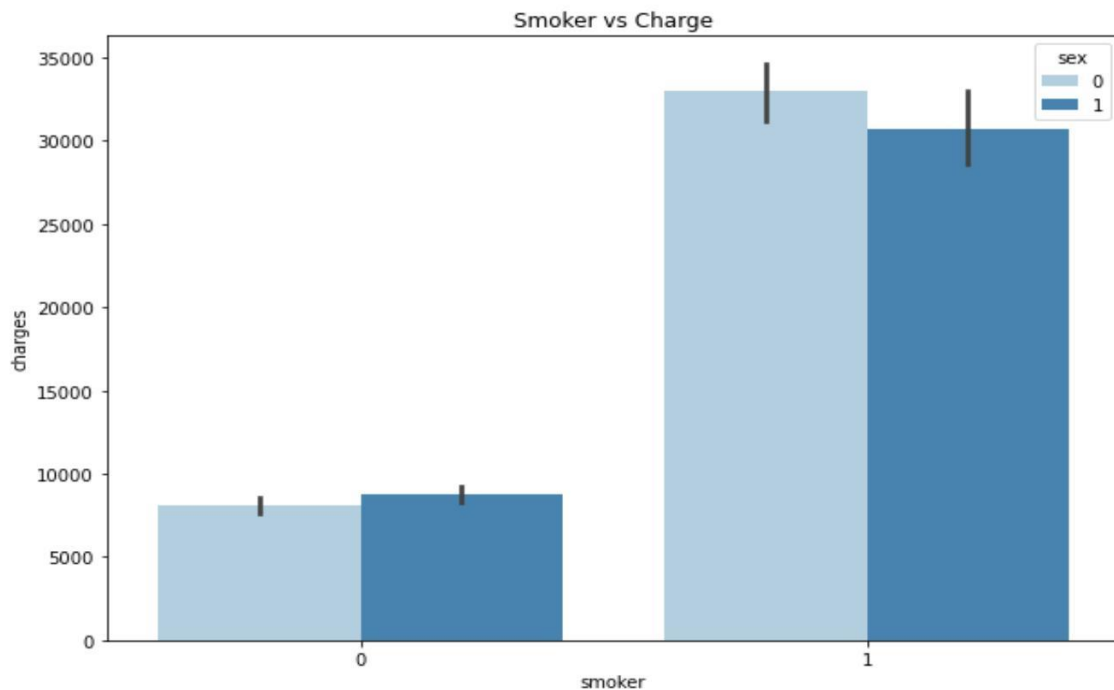


In this figure, the females are represented by a value of zero, while the men are represented by a value of one. The x-axis displays the BMI values for the two genders (male and female), while the y-axis displays the charges. It is obvious that when the BMI values are changed, the insurance costs would change as well, as illustrated in Figure.

Smoker vs Charge

```
plt.figure(figsize=(10,7)) plt.title('Smoker  
vs Charge')  
sns.barplot(x='smoker',y='charges',data=data_copy,palette='Blues',hue='sex')
```

Out[17]: <AxesSubplot: title={'center': 'Smoker vs Charge'}, xlabel='smoker', ylabel='charges'>

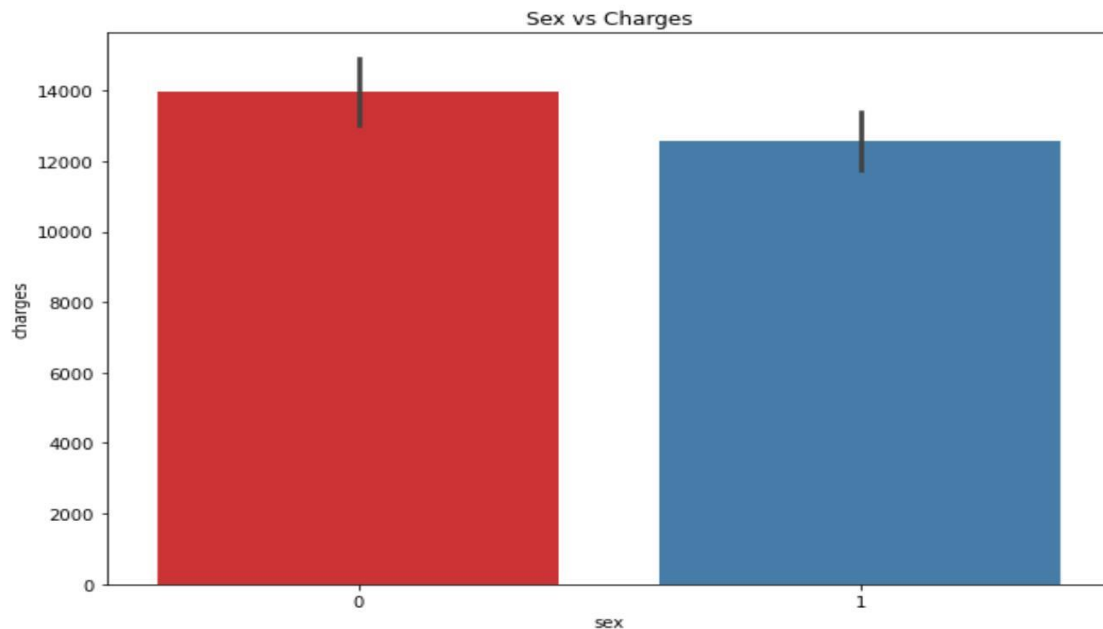


The Figure shows how the cost of medical insurance varies somewhat for an average smoker. The cost of health insurance is higher for women than for men since men are more addicted to and enthusiastic about smoking than women. Figure shows us that when smoking rates rise, insurance premiums for males would decline while those for women will rise. The x-axis displays the values of smokers, while the y-axis displays the charges.

Sex vs. Charges

```
plt.figure(figsize=(10,7)) plt.title('Sex  
vs Charges')  
sns.barplot(x='sex',y='charges',data=data_copy,palette='Set1')
```

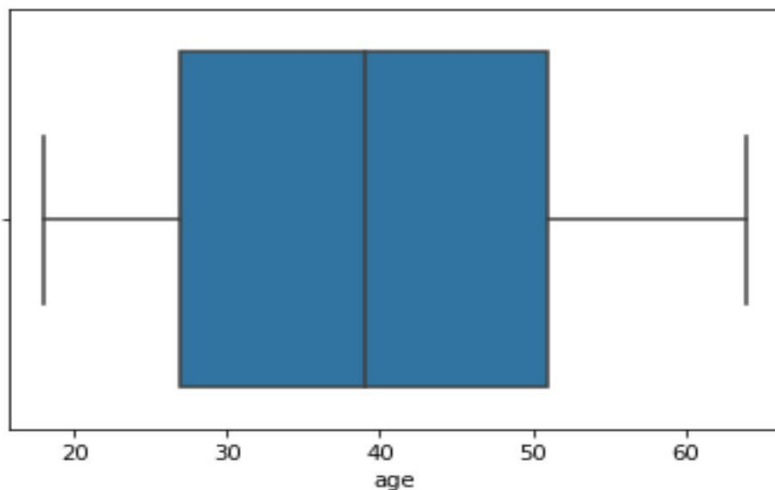
Out[18]: <AxesSubplot: title={'center': 'Sex vs Charges'}, xlabel='sex', ylabel='charges'>



According to Figure, the cost of medical insurance is always higher for women than for men. The sex kinds are listed on the x-axis, and the costs are shown on the y-axis. The graph shows that the female insurance costs 14,000, while the male insurance costs are closer to 13,000.

Boxplot

```
for i in range(0,6):
sns.boxplot(data_copy.iloc[:,i]) plt.show()
```



Feature Scaling:

A technique for normalising the variety of independent variables or features in data is called feature scaling. It is often carried out during the data preparation stage and is sometimes referred to as data normalisation in the context of data processing.

There are 2 Methods of feature scaling:

1. Normalisation: The process of normalisation, often referred to as min-max scaling, involves shifting the numbers in a column such that they are contained inside a specified range between 0 and 1.
The normalisation Scikit-learn function is called Minmax Scaler.
2. Standardisation: However, standardisation, also known as Z-score normalisation, is another scaling approach that involves rescaling the values in a column to reflect the characteristics of a conventional Gaussian distribution, which has a mean of 0 and a variance of 1.

The Scikit-learn function for standardisation is called StandardScaler.

```
from sklearn.preprocessing import StandardScaler
data_pre = data_copy.copy()

tempBmi = data_pre.bmi
tempBmi = tempBmi.values.reshape(-1,1) data_pre['bmi']
= StandardScaler().fit_transform(tempBmi)

tempAge = data_pre.age
tempAge = tempAge.values.reshape(-1,1) data_pre['age']
= StandardScaler().fit_transform(tempAge)

tempCharges = data_pre.charges
tempCharges = tempCharges.values.reshape(-1,1)
data_pre['charges'] = StandardScaler().fit_transform(tempCharges)

data_pre.head()
```

	age	sex	bmi	children	smoker	region	charges
0	-1.438764	1	-0.453320	0	1	3	0.298584
1	-1.509965	0	0.509621	1	0	2	-0.953689
2	-0.797954	0	0.383307	3	0	2	-0.728675
3	-0.441948	0	-1.305531	0	0	0	0.719843
4	-0.513149	0	-0.292556	0	0	0	-0.776802

Assumption of Linear Regression:

1. Assumption 1 — Independence of observations

The independence of the data is the first assumption in linear regression. Independence denotes the absence of any connection between the several samples. The technique used to acquire the data is more likely to provide an answer to this than just looking at the data can.

2. Assumption 2 — No Hidden or Missing Variables

The second premise of a linear regression model is that all pertinent explanatory variables have been included in our model. If we don't, the model will attempt to assign coefficients to the variables that are present in our data set, which will result in an incorrect model. This is frequently referred to as model misspecification.

If including a variable in the model will significantly alter the results, the model is flawed and unusable without it. In this situation, our only option is to return to our data collection and locate the required information.

3. Assumption 3 — Linear relationship

The third tenet of linear regression is that there must be linear relationships between the independent and dependent variables.

Even though this presumption is not often mentioned in the literature, it makes sense and should be verified. Since our relationships are not linear, we should instead use a non-linear model, of which there are several available.

4. Assumption 4 — Normality of the Residual

The residuals should adhere to a normal distribution, which is the fourth premise of linear regression. Once we have our model's residuals, it is rather simple to assess this using a histogram or a QQ Plot. Although QQ Plots are more difficult to read, they are easier to comprehend.

5. Assumption 5 — No or little Multicollinearity

There is either no or very little multicollinearity, which is the sixth assumption of linear regression. Multicollinearity is a phenomenon that occurs when many explanatory variables have a high degree of correlation.

Why then do we desire zero correlation across independent variables yet substantial correlations between each independent variable and the dependent variable? Because they both provide an explanation for the same data, two independent variables that are associated. The dependent variable will change, but the model won't be able to determine which of the two variables is truly to blame.

The Variance Inflation Factor, or VIF for short, can be used to check for multicollinearity issues. The VIF shows the degree to which one independent variable is associated with the other independent variables.

6. Assumption 6 — Homoscedasticity

Homoscedasticity is the sixth presumption of linear regression. When a model is homoscedastic, the error is constant throughout the range of values for the dependent variable. The easiest technique to determine homoscedasticity is to plot the residuals against the dependent variable in a scatterplot.

7. Assumption 7 — All independent variables are uncorrelated with the error term

The linear regression model's seventh diagnostic check examines the relationship between the independent variables and the error term. If this occurs, it probably has a model that was incorrectly stated. We could have overlooked a crucial explanatory factor.

8. Assumption 8 — Observations of the error term are uncorrelated with each other

The last assumption of the linear regression model states that the error term should be uncorrelated with each other.

Data Preprocessing:

Split dataset in Train and Test:

In data science and machine learning, the train-test split approach is used to assess how well a prediction model is doing. The given data is separated into two distinct sets for train-test split: a training set and a testing set. The testing set is used to assess how well the model performs when applied to new data, whereas the training set is used to train the model.

By modifying the model's parameters depending on the input data and the associated output, the training set is used to fit the model to the data. The performance of the model on fresh, untested data is then assessed using the testing set. Metrics including accuracy, precision, recall, and F1 are commonly calculated to assess the model's performance.

Overfitting, which happens when a model is too complicated and can match the training data too closely, is something that the train-test split aims to prevent. When a model is used to analyse fresh, unknown data, overfitting can cause poor performance. It is feasible to determine if the model is overfitting or underfitting and change the model's parameters accordingly by assessing the model's performance on a different testing set.

Several methods, including random sampling, stratified sampling, and k-fold cross-validation, can be used to conduct the train-test split. The quantity and makeup of the dataset, as well as the demands of the modelling assignment, all influence the approach chosen.

```
X = data_pre.drop('charges',axis=1).values y
= data_pre['charges'].values.reshape(-1,1)

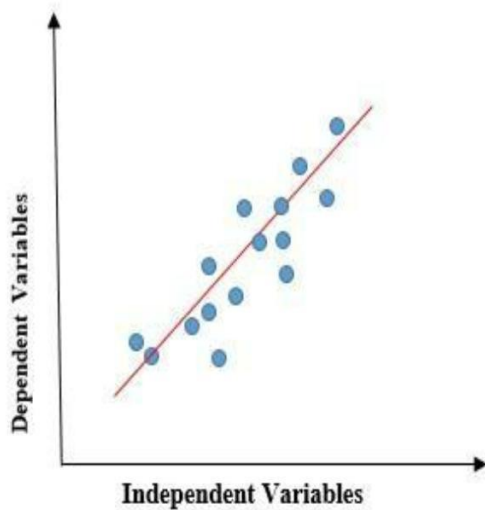
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2, random_state=42)

print('Size of X_train : ', X_train.shape) print('Size
of y_train : ', y_train.shape)
print('Size of X_test : ', X_test.shape)
print('Size of Y_test : ', y_test.shape)
```

Modelling:

1. Linear Regression

A modest and straightforward statistical regression technique called linear regression is utilised to highlight the link between continuous variables in predictive analysis. The term "linear regression" refers to a statistical method that displays a linear connection between the independent variable (X-axis) and the dependent variable (Y-axis). Such linear regression is known as simple linear regression if there is just one input variable (x). Additionally, this type of linear regression is known as multiple linear regression if there are many input variables. The link between the variables is depicted by a sloping straight line in the linear regression model.



The dependent variable's linear connection with the independent variables is seen in the graph above. When the value of x , an independent variable, rises, so does the value of y , a dependent variable. The best suited straight line is denoted by the red line. We attempt to draw a line from the provided data points that best represents the points. To calculate best-fit line linear regression uses a traditional slope-intercept form.

Linear Regression equation

$$y = mx + b \implies y = a_0 + a_1x$$

$y =$

Dependent Variable.

$x =$ Independent Variable.

$a_0 =$ intercept of the line.

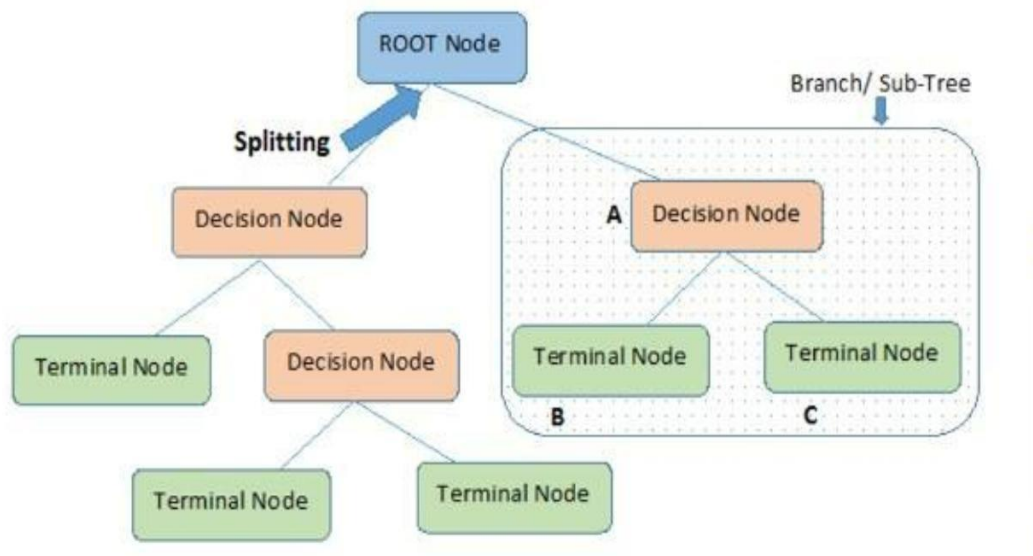
$a_1 =$ Linear regression coefficient

.

2. Decision Tree Regression

A hierarchical model known as a decision tree is used in decision support to represent decisions and their potential consequences, including chance events, resource costs, and utility. This algorithmic approach, which is non-parametric, supervised learning, and appropriate for both classification and regression applications, makes use of conditional control statements. A root node, branches, internal nodes, and leaf nodes make up the tree structure, which has a hierarchical tree-like shape.

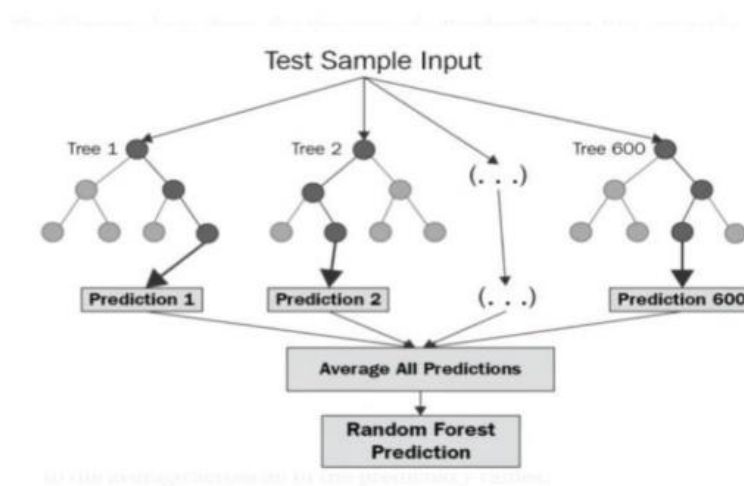
It is a tool with uses in several different fields. Both classification and regression issues may be solved using decision trees. The term itself implies that it displays the predictions that come from a sequence of feature-based splits using a flowchart that resembles a tree structure. The choice is made by the leaves at the end, which follows the root node.



3. Random Forest Regression

A supervised learning technique called Random Forest Regression leverages the ensemble learning approach for regression. The ensemble learning approach combines predictions from many machine learning algorithms to provide predictions that are more accurate than those from a single model.

The structure of a Random Forest is seen in the diagram above. The parallel alignment of the trees and their complete lack of interaction are obvious. During training, a Random Forest builds many decision trees and outputs the mean of the classes as the forecast of all the trees.



The structure of a Random Forest is seen in the diagram above. The parallel alignment of the trees and their complete lack of interaction are obvious. During training, a Random Forest builds many decision trees and outputs the mean of the classes as the forecast of all the trees.

4. Support Vector Regression

The regression technique known as Support Vector Regression, as its name indicates, enables both linear and non-linear regressions. The Support Vector Machine is the basis for how this approach operates. In contrast to SVM, which is used to predict discrete categorical labels, SVR is a regressor that is used to forecast continuous ordered variables. This is one-way SVR and SVM vary from one another.

The goal of basic regression is to reduce error rates, but the goal of SVR is to fit the error within a predetermined threshold. This implies that the task of SVR is to approximate the best value within a predetermined margin known as the ϵ -tube. See the following illustration to comprehend SVR:

- A hyperplane is a boundary between two data classes in a dimension higher than the real dimension. It is described in SVR as the line that aids in forecasting the desired value.
- Kernel: In SVR, a higher dimension is used for the regression. We require a function to map the data points into the higher dimension in order to do that. The kernel is the name given to this function. Sigmoidal Kernels, Polynomial Kernels, Gaussian Kernels, etc. are the types of kernels utilized in SVR.
- Boundary Lines: These are the two lines that are drawn at a distance of (ϵ) around the hyperplane. It serves as a buffer between the data points.
- Support Vector: The vector used to define the hyperplane, or alternatively, the extreme data points in the dataset that aid in creating the hyperplane. The border is close to these data points.
- SVR seeks to accommodate as many data points as it can while staying inside the margin. Be aware that whereas support vectors were used to build the hyperplane in SVM classification, they were utilized to define the linear regression in SVR.

5.Gradient Boosting Algorithm

A machine learning method called gradient boosting is used, among other things, for classification and regression problems. It provides a prediction model in the form of an ensemble of decision trees-like weak prediction models. The resultant technique, known as gradient-boosted trees, typically beats random forest when a decision tree is the weak learner. The construction of a gradient-boosted trees model follows the same stage-wise process as previous boosting techniques, but it generalizes other techniques by enabling the optimization of any differentiable loss function.

Conclusion:

In this paper, by using a set of ML algorithms, a computational intelligence approach is applied to predict healthcare insurance costs. The medical insurance dataset was obtained from the KAGGLE repository and was utilized for training and testing the Linear Regression, Decision Tree Regressor ML algorithms. The regression of this dataset followed the steps of pre-processing, feature engineering, data splitting, regression, and evaluation. The resultant outcome revealed that Linear Regression achieved a high accuracy of 87.9%.